Databricks Project

BootCamp_Project-2

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Program: Azure Data Engineering

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1. Project Overview

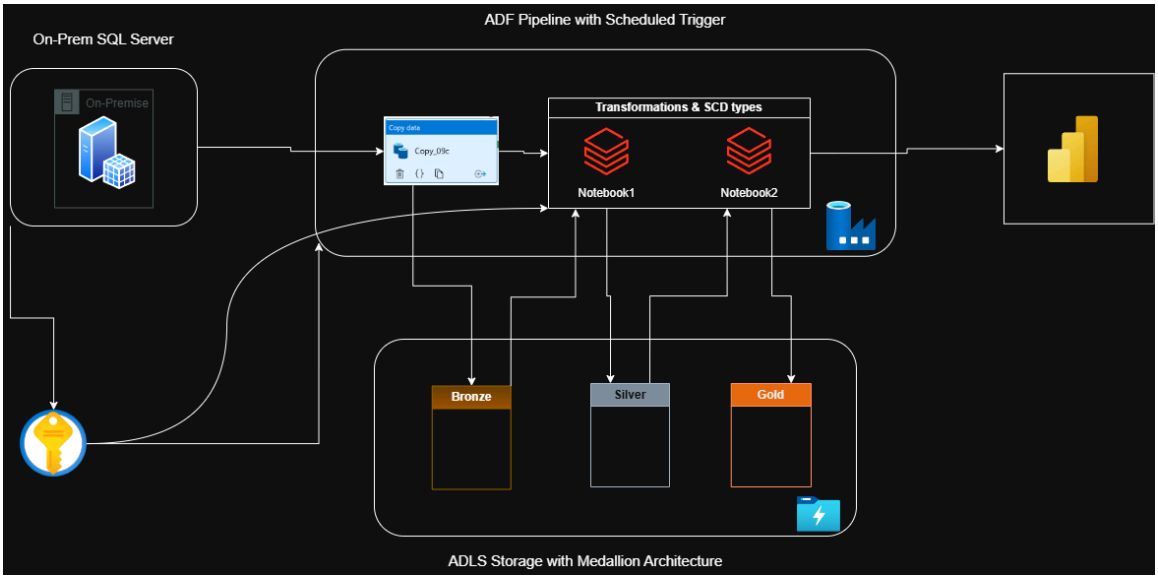
This project demonstrates the implementation of a scalable end-to-end data engineering pipeline using Azure services. The pipeline ingests data from an on-premises SQL Server, processes it using Azure Databricks, and delivers business insights through Power BI.

The architecture follows the Medallion Architecture (Bronze, Silver, Gold layers) to ensure data quality, scalability, and maintainability.

2. Objectives

- Extracting data from on-prem SQL Server using ADF
- Store raw data in ADLS Gen2 (Bronze layer)
- Perform data cleaning and transformation using PySpark (Silver layer)
- Apply SCD Type 1 for dimension management (Gold layer)
- Generate business insights like revenue, profit, and customer trends
- Visualize data using Power BI

3. Architecture Diagram



4. Technology Stack

- Azure Data Factory (ADF) – Data ingestion
- Azure Data Lake Storage Gen2 (ADLS) – Storage
- Azure Databricks – Data transformation
- PySpark – Data processing
- Delta Lake – Storage format
- Power BI – Visualization

5. Data Model (E-commerce)

Tables Used:

- Customers
- Products
- Orders
- OrderItems

- Payments

Results | Messages

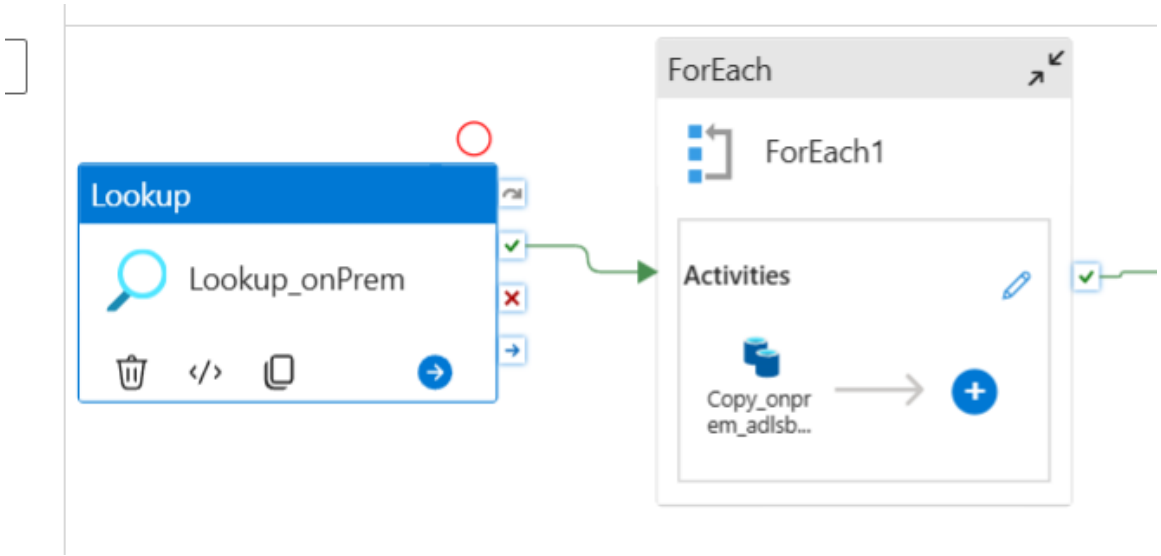
	TABLE_CATALOG	TABLE_SCHEMA	TABLE_NAME	TABLE_TYPE
1	ADE_trainingdb	dbo	Customers	BASE TABLE
2	ADE_trainingdb	dbo	OrderItems	BASE TABLE
3	ADE_trainingdb	dbo	Orders	BASE TABLE
4	ADE_trainingdb	dbo	Payments	BASE TABLE
5	ADE_trainingdb	dbo	Products	BASE TABLE

6. Data Ingestion (ADF - ADLS Bronze Layer)

Data is extracted dynamically from SQL Server using an ADF pipeline.

Pipeline Activities:

- Lookup → Fetch table names
- ForEach → Iterate tables
- Copy Activity → Load into ADLS



7. Data Transformation (Databricks - Silver Layer)

Data from the Bronze layer is processed using PySpark notebooks.

Transformations Performed:

- Removal of duplicate records
- Null value filtering
- Date format standardization
- Column renaming
- Adding ingestion timestamp
- Removing unwanted columns

Example Transformation:

```
df_clean = df.dropDuplicates()

df_clean = df_clean.filter(col("CustomerID").isNotNull())

df_clean = df_clean.withColumn("CreatedDate", to_timestamp("CreatedDate"))

df_clean = df_clean.withColumn("ingestion_time", current_timestamp())
```

Output:

Storing the output delta tables to Silver layer.

8. Business Logic Implementation (Gold Layer)

A separate notebook creates analytical datasets.

Metrics Created:

- Revenue = Quantity × Selling Price
- Profit = (Selling Price – Cost) × Quantity

Output Table:

gold.sales_metrics

9. Analytical Insights

The following insights are generated:

- Highest cost product
- Customer with highest purchase quantity
- Total revenue per product
- Profit distribution

These are calculated using PySpark aggregations.

10. SCD Type 1 Implementation

SCD Type 1 is applied to maintain the latest customer data.

Logic:

- If record exists → Update
- If not → Insert

Implementation:

```
if not DeltaTable.isDeltaTable(spark, scd_path):   #checking the
existence of the table if not it will create table



    df_init = df_customer \

        .withColumn("StartDate", current_date()) \

        .withColumn("EndDate", lit(None).cast("date")) \

        .withColumn("IsCurrent", lit(1))



    df_init.write.format("delta").mode("overwrite").save(scd_path)
```

```
else:
    delta_table = DeltaTable.forPath(spark, scd_path) # loading the table


    delta_table.alias("t").merge(
        df_customer.alias("s"),
        "t.CustomerID = s.CustomerID AND t.IsCurrent = 1"
    ).whenMatchedUpdate(set={
        "FirstName": "s.FirstName",
        "LastName": "s.LastName",
        "Email": "s.Email",
        "City": "s.City"
    }).whenNotMatchedInsertAll().execute() # condition for matching
current records
```

Output:

gold.customers_dim

11. Data Storage Strategy

- Bronze → Raw CSV files
- Silver → Cleaned Delta tables
- Gold → Business-level Delta tables

12. Power BI Integration

Steps:

1. Connect to Azure Databricks
2. Select Gold tables
3. Build dashboards:
    - Revenue by product
    - Profit analysis
    - Top customers

13. Security Implementation

- Azure Key Vault used for storing credentials
- Secure access via Service Principal
- No hardcoded secrets in notebooks

14. Error Handling

- Null checks implemented during transformation
- Schema validation before processing
- Duplicate handling using dropDuplicates()
- Merge conflicts handled using Delta Lake

15. Challenges Faced

Handling missing headers in CSV files caused schema issues during transformation. This was resolved by enabling header options and validating schema before processing.

Another challenge was managing schema mismatches between source and target tables during SCD implementation. This was fixed by aligning column structures and ensuring consistent schemas.

Performance optimization was required when processing large datasets. This was addressed by using Delta Lake optimizations and efficient transformations in PySpark.

16. Performance Optimization

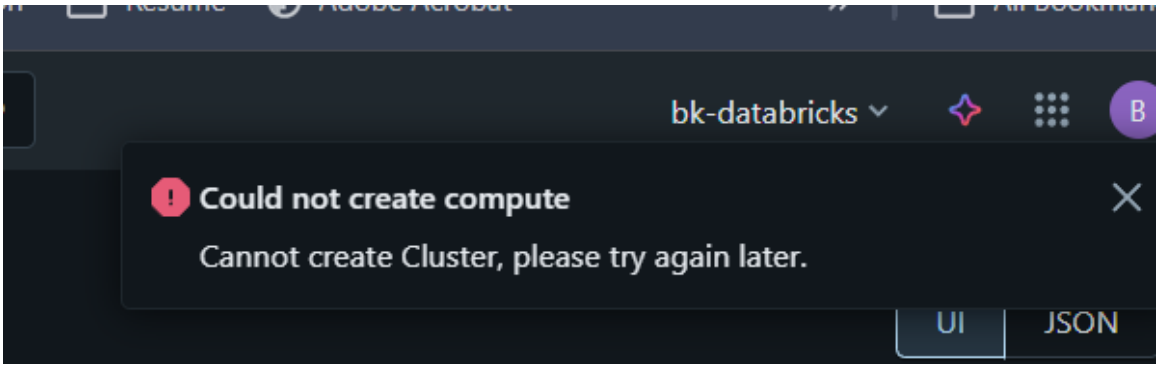
- Used Delta Lake for faster reads/writes
- Applied partitioning (optional)
- Used caching where required
- Optimized queries using aggregation functions

17. How to Run the Project

1. Create tables in SQL Server
2. Run ADF pipeline to load data into ADLS Bronze
3. Execute Databricks connection notebook
4. Run transformation notebooks (Silver layer)
5. Run business logic notebook
6. Execute SCD Type 1 notebook
7. Connect Power BI and build dashboard

18. Conclusion

This project successfully demonstrates a modern data engineering workflow using Azure services. It highlights best practices such as layered architecture, scalable data processing, and efficient data modeling. The solution is capable of handling large datasets and delivering business insights in a structured and reliable manner.


bk-databricks
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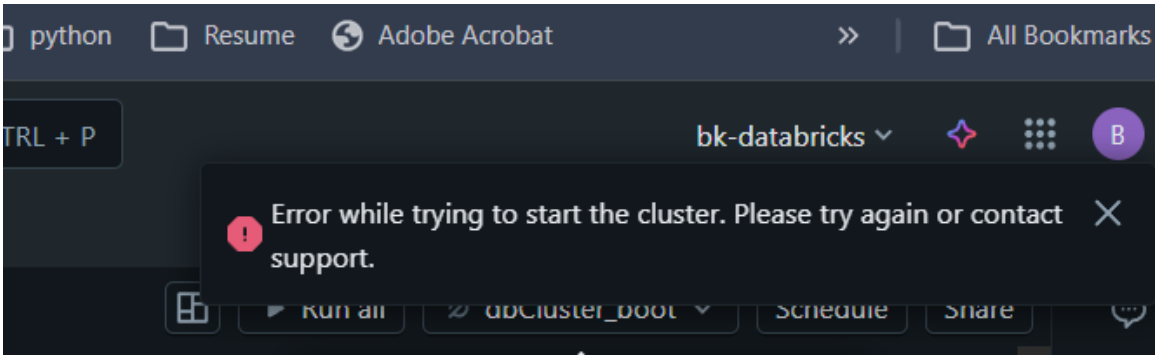
Could not create compute
Cannot create Cluster, please try again later.
UI
JSON


python
Resume
Adobe Acrobat
All Bookmarks
TRL + P
bk-databricks
B
Error while trying to start the cluster. Please try again or contact support.